Rethinking Assignments and AI

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Introduction

Generative AI technology raises at least two kinds of questions about the way faculty design assignments and learning experiences for their students. One set of questions concerns what new things students might need to learn in a world where AI tools are proliferating.

- Is there background information about AI they'll need, or new skills they'll need to learn?
- If so, how might we design assignments that help make that context vivid or put them in a position to practice those skills?

Another set of questions concerns how students can demonstrate what they've learned.

- If an AI tool can produce text (or a poster, PowerPoint deck, video clip, or podcast recording) that looks or sounds human, can we continue to rely on products like these as evidence of what students know or can do?
- If so, how? And if not, what should we do instead?

In the summer and fall of 2024, the Center for Teaching and Learning hosted a series of workshops in which faculty could consider these questions together. This document collects some of the upshots from those conversations. The sections below correspond to the workshops in that series:

- one providing background information about generative AI,
- one about designing assignments to promote AI literacy,
- one about discouraging dishonest or inappropriate uses of AI,

- and one about making sure that you and your students are on the same page about these matters.

Our hope is that collecting those reflections here provides a helpful refresher for those who could participate in those workshops and a helpful primer for those who couldn't. In each case, however, we're keenly aware that those workshops were the start of a conversation rather than the end of one. So, we encourage you to read this document with a critical eye and to seek out other resources where they're applicable. For more information, you might consider:

- the Library's subject guide,
- this list of recommended reading the CTL put together to accompany the workshop series,
- the CTL's suggestions about syllabus language,
- the student-facing guide to AI put together by Elon University and the AAC&U,
- or reaching out to us to talk through the thoughts that emerge for you as you read (you can find us at ctl@pacific.edu).

Getting Started with Generative AI

To inform your thinking about what generative AI means for your courses, it can be helpful to have some working knowledge about how it works and what it can do. What makes these tools "artificial intelligence" is that they are computer systems that mimic human intelligence in their outputs. What makes them "generative" is that these outputs go beyond performing predetermined operations on the inputs humans give them, so that users (and developers) are not in a position to predict exactly what they'll come up with. For example, chatbots (like ChatGPT) can generate text that sounds human-written and image generators (like DALL-E) can produce graphics that look human-designed. Those tools are intended to be general purpose – you could talk with a chatbot about just about anything – but there's a large and growing array of tools built on this technology that are tailored to more specific purposes. For example, the writing assistant Grammarly can suggest revisions for tone or style and the research aid ResearchRabbit can create a citation map representing the sources that are driving the conversation in the academic literature on a particular topic.

All of these tools are built on "Large Language Models" (commonly called "LLMs"). These are systems that can take in large amounts of data, use mathematical operations to identify patterns in that data, then use those patterns to make predictions about what a human-like response to a new prompt would be. The computational details are beyond the scope of this primer, but it can be helpful to have a metaphor for what these programs do. Computer scientists often describe LLMs as "stochastic parrots." They're parrots in the sense

that they do not work by understanding the inputs we give them, reasoning about those inputs, and arriving at a result; instead, they work by parroting something that sounds like what a human might say or do. They're stochastic in the sense that there is unpredictability in how they respond; given the same input on two different occasions, a chatbot might produce two different responses.

This way of thinking about what generative AI tools are doing helps to illuminate some of what makes them useful and some of their potential pitfalls. The fact that generative AI doesn't understand what it's saying can be leveraged for the purposes of brainstorming. If you ask me to come up with 50 different possible titles for your new book project, I'll run out of ideas after four or five. Unencumbered by a background understanding of what makes something a good title (or worries about what you'll think of its work), though, a chatbot can keep generating things that sound title-ish more or less endlessly. You can then pick the one that works for your purposes, or take bits that you like from several and use your actual understanding of your project to combine them. This ability to produce possibilities free of humans' prior assumptions can be a helpful stimulus for human creativity.

That generative AI works by identifying patterns can also be helpful when you yourself are in a situation that calls for emulating a particular style or tone, particularly when it doesn't come easily to you. For example, imagine that you know about yourself that you tend to be curt in emails. You might then ask a chatbot to rewrite your email with more warmth. Of course, it won't understand the content of your email, so some of its suggestions might not be appropriate. However, seeing how it mimics the type of writing you're going for can give you a useful model to work from.

These same features of how generative AI works illuminate some of the ways it can go wrong. Given that generative AI tools don't understand what they're saying, they're prone to producing results that sound plausible but are inaccurate. This phenomenon is sometimes called "hallucinating." For example, when asked to write a literature review, a generative AI tool might predict that what appears next will have the form of a citation. But, without understanding what citations are or their role in scholarly practice, it won't know to fill in the citation with a source that really exists. If you're an expert in the field this literature review is about, you'll probably notice that something seems amiss; if you're a novice, you might not. So, using these tools responsibly requires some self-knowledge on the part of the user.

Another pitfall stems from the fact that generative AI works by identifying patterns in the data that it's trained on. When that data is incomplete, misleading, or morally objectionable, the AI's outputs will be too. This phenomenon is sometimes called "algorithmic bias." For example, given existing gender inequity in our society, doctors are disproportionately men. As a result, when asked to produce a picture of a doctor, an image generator is disproportionately likely to produce a picture of a man. Similarly, AI summaries tend to reflect the dominant approach to a topic, and as a result will be likely to miss under-discussed views or

contributions from authors with minoritized identities. So, uncritical use of these tools risks reinforcing inequities that shape the data they're trained on.

To get a feel for what these tools can do and what their limitations are, it can be helpful to try them for yourself. If you'd like to do that, here's a useful exercise. First, choose a task that's routine for you, but requires some specialized knowledge. For example, you might choose drafting a lesson plan, a case study relevant to a course you teach, an assignment prompt that's relevant to students' interests, or a series of exam questions. Next, try prompting an AI tool (or several of them, to compare the results) to do that task several times.[1] For example, you might ask for three versions of a lesson plan, or one case study for each student in your class. Finally, reflect on the results you get. Do they give you any ideas you wouldn't have come up with on your own? Are there aspects of this task you tend to get stuck on, where looking to these could help? Do you spot any hallucinations? Are there nuances that are missing or popular misconceptions that are reproduced? Playing around with the tools in this way can help you get a richer sense of how you might use them and what options your students might be considering.

If you try this, one thing you're likely to find is that AI tools tend to do better when given more direction, rather than less. Doing things like providing an identity or context ("you are a professor who teaches an introductory level course on…"), providing constraints ("the questions should have four options and exactly one correct answer…"), and providing an audience ("the questions should be an appropriate difficulty level for students in their first semester of college…") can all improve the responses you get. Prompting well is a skill, sometimes called "prompt engineering."

There is a lot to be learned from experimenting with these tools yourself, but there are also a variety of reasons you might consider abstaining. For example, AI tools are resource-intensive, so you might limit your use if you are concerned about your carbon footprint. You might also be concerned about the labor conditions for people who test these tools, about the possibility that they'll be used for scams and that human-sounding fakes might erode trust in media or public institutions, about how the data you feed into them when prompting will be used, about how these tools complicate questions about authorship, or about the way that outsourcing decisions to algorithms might allow corporations or governments to dodge accountability.[2] You may find these compelling reasons, and so may many of your students. We recommend taking some time to reflect on your own views, as well as keeping in mind that your students will probably come into your courses with different levels of familiarity with these issues and different levels of comfort with AI.

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[1] For a sense of some tools you might consider starting with, here are some options recommended in the Library Subject Guide.

[2] For more detail about these concerns, see this overview in the Library Subject Guide.

Designing Assignments to Promote AI Literacy

There are a variety of reasons that you might want students to learn something about AI in your courses. You might see ways that AI tools are changing your field, you might want to prepare students to use AI in their work lives, you might think that understanding how AI works will contribute to students' information literacy, or that understanding the various moral questions it raises will contribute to their ability to participate meaningfully in public life. More pragmatically, you might just think that some students are likely to use AI when engaging with your course and decide that if they're going to use it, you want to equip them to use it thoughtfully.

In light of these different kinds of motivations, the first step when designing assignments that promote AI literacy is deciding what, exactly, you want students to know, feel, or be able to do with respect to AI. Are you hoping that they'll be able to use a particular prompt engineering strategy? Evaluate AI output for hallucinations or algorithmic bias? Become self-aware about their own writing process and where AI fits into it? Each of these goals might be served by different assignments or activities.

Once you have your goals in mind, assignments that promote AI literacy tend to have a common form. You start by asking students to do something with AI (or by providing them with something done by AI) and then reflecting on that AI output in some way. You can tailor this structure to your particular goals by varying what you're having AI do and what the focus of students' reflection is. For example, if your goal is to help students identify hallucinations, you might ask them to use what they've learned in class to evaluate an AI's attempt to answer questions about course topics. If your goal is to help students identify algorithmic bias, you might have them ask an AI about a topic they know a lot about and take note of ways they think its answers reflect dominant perspectives on that topic. If your goal is to develop students' prompt engineering skills, you might ask them to try different prompt engineering strategies and reflect on what differences in AI outputs they notice.

In the examples above, you're asking students to complete an assignment specifically geared toward learning something about AI. It's also possible to add an AI literacy element to an assignment that plays some other role in your course. For example, imagine that you have students making a presentation, collaborating on programming code, or writing a lab report. In addition to whatever they would normally turn in to you after doing that, you might ask them to submit some commentary about how they used AI. As in the other examples, you can tailor the focus of students' commentary to suit your goals. If you're focused on prompt engineering, you might ask them why they used the particular prompts they did. If you're focused on questions about authorship, you might ask them what they did to ensure that the final product was their own.

As you decide how to adapt these strategies to your courses, the CTL recommends keeping three further points in mind. First, it can be helpful to think about different places in the process of working on an assignment that students might use AI. In the context of a research paper for example, students might use it to brainstorm topics, to find relevant sources, to generate outlines or text, to anticipate questions or challenges from different perspectives, or to get feedback on text they've written. Given this variety, students might be well-served by thinking not just about how to use AI but also when to use it. One way to accomplish this is to have students try using AI at different stages for different assignments, then compare and contrast. It may turn out that when students do this, they find that there are individual differences in their experiences. For example, some students might find that using AI for brainstorming gives them a place to start, so that their own thinking can happen in revising or remixing what AI offers them. Other students, in contrast, might find that the brainstorming phase is where they do most of their thinking, so that using AI for that purpose tends to crowd out their own intellectual contributions. In light of these differences, using AI well might require self-knowledge about one's work process in addition to knowledge about AI.

You might also consider building AI into the process by which students get feedback on their work. Given that it doesn't understand the content, the feedback won't always point students in the right direction. However, given a well-constructed prompt, it can often do a surprisingly good job (here is a library of prompts illustrating different kinds of feedback students might request). Feedback from AI has a faster turnaround time than faculty are often in a position to offer and is available at times when faculty are not, making it a potentially useful supplement to (but not a replacement for) feedback students get from you. If you do decide to invite students to use AI in this way, it can be helpful to give them some structure for assessing the feedback they get and deciding what to do with it. This can help them avoid uncritically accepting AI suggestions and can reinforce their sense of ownership of their work (here are some examples of templates students can use for this purpose – these examples are about feedback on writing, but they could be adapted for feedback on other kinds of student work).

Second, when designing assignments with an AI literacy component, we recommend prioritizing among your goals. There's a lot it might be good for students to know about AI, so it can be tempting to ask them to reflect on many aspects of their AI use all at once. Adding complexity risks overwhelming students, preventing them from focusing their attention on the goals of the assignment that are most important to you. When there are many takeaways you want students to have, you might consider breaking them up over the course of several smaller assignments, or building up from less to more complex tasks.

Finally, it's worth remembering that just as individual faculty might have moral reasons for limiting their AI use, individual students may have principled concerns as well. So, when thinking about promoting AI literacy, it's worth considering what you can do to accommodate these students. Do the learning outcomes of your course absolutely require AI use? Or are

there alternatives, like discussing others' experiences with AI or engaging with examples provided by the instructor? Whether or not accommodation is possible will depend on the context of your course, so there is more than one reasonable answer here. Nonetheless, it's worth keeping in mind that students' prior familiarity with AI and willingness to use it will vary.

Designing Assignments to Motivate Human Effort

Just as there are many reasons you might want students to learn about using AI well, there are many reasons you might want students not to use AI for particular tasks, or not to use it in particular ways. One fundamental reason is that often what's important about the assignments we give students is the process students go through when working on them: recalling information, organizing and making sense of it, practicing relevant skills, and so on. Copying and pasting an AI output skips over that process, so the assignment no longer serves as an opportunity to learn. There are other, more pragmatic reasons as well. You might be preparing students for a setting in which AI support is not readily available (like a clinical practice) or you might need students to develop sufficient skill and content knowledge to critically evaluate what AI comes up with.

As a first step toward thinking about how to approach situations where you do not want students to use AI, it's helpful to reflect on your goals for the particular assignments that you want students to do without AI assistance. Doing so can help you determine exactly what kinds of AI use are incompatible with those goals. For example, if you ask students to write lab reports to show their understanding of the scientific principles at work in their lab experiments, then using AI to write explanations of those principles would undermine the purpose of the assignment, but using AI to correct grammar would not. In contrast, if you ask students to write personal reflections in a language they're learning as a way to practice their grammar, then using AI to correct grammar would undermine the purpose of the assignment. In short, what counts as an inappropriate use depends on your goals.

Once you have your goals clearly in mind, you can think about how to discourage types of AI use that are incompatible with those goals. There are a number of strategies the CTL suggests for how to do this, but when choosing among them it can be helpful to keep two preliminary points in mind.

First, there are a wide range of reasons that students might be tempted to use AI inappropriately. Here, for instance, is a sampling of the reasons faculty came up with during the workshop series:

- They might be overwhelmed by other coursework or other responsibilities;
- They might not understand the value of what they're being asked to do;

- They might be worried that they're not capable of doing what they're being asked to do or might not know where to start;
- They might think that AI-generated text sounds smarter or more academic than their writing does;
- They might not be sure which uses are appropriate and which aren't;
- They might be worried about keeping up with classmates who are doing it.

This range suggests that the most effective ways of motivating students' human effort are going to be context dependent. For example, strategies for helping students appreciate the value of a task might look different from strategies for helping students see themselves as capable of that task. So, finding out about your students' thinking about your courses can help you choose to the right mix of strategies for your context.

Second, the possibility of students cheating is not new. Prior to the widespread availability of AI, students could look up homework answers on repositories like Chegg, seek answers from peers, or pay essay mills to do their work for them. This comparison serves as a reminder that assessment security – our confidence that students' work represents their learning – is not the only relevant variable in assignment design. Sometimes making an assessment more secure also makes it less authentic, as when an in-class essay doesn't leave time for the kind of reflection and revision that are part of writing as it is actually practiced in your discipline. Sometimes making an assessment more secure also makes it less equitable, as when adding a timer to an online quiz disproportionately affects students with dyslexia. And sometimes making an assessment more secure undermines your relationship with your students, promoting the idea that attempting to cheat is just part of the academic game. This is not to say that you should not discourage students' inappropriate uses of AI, just that in doing so you should not lose sight of your other goals.

How might you do that? One approach is to reinforce students' motivation to do the task you're asking them to do. Strategies aligned with this approach include:

- Explaining how doing an assignment will help students (e.g. what skills they'll practice or how they'll use those skills in other contexts) can help students see value in a task. Alternatively, you can invite students to identify the value themselves as part of a class discussion.

- Designing tasks that are relevant to students' lives or outside interests or prompting students to identify connections between course content and contexts outside of school can tap into their intrinsic motivation.

- Giving students options so that they can focus on the aspects of a topic that most interest them, or on presenting their work in ways that are meaningful and accessible

can boost students' sense of self-efficacy.

- Providing frequent, low-stakes opportunities for practice can alleviate students' anxiety about making a mistake.

- Normalizing mistakes and emphasizing the importance of adjusting based on feedback can help students feel less fearful about taking intellectual risks. For example, you might have conversations about your own work in your discipline or about perseverance by people in your field.

- Explicitly connecting what students do in class to assignments that students will complete on their own later can help ensure that students feel prepared when it comes time to work on their own.

- Working in groups can help students feel more connected to the campus community. It can also create a sense of accountability to their groupmates. This is particularly effective when you support groups in thinking about what makes a group work well together, establishing expectations for group members, and thinking in advance about how they'll address potential problems.

- Co-creating policies with students can give them a sense of autonomy, increasing their motivation to abide by the policy.

- When students' work will be seen or used for something outside of class, it reinforces their sense that what they do matters. This can be done in small ways, like students presenting what they've learned at a poster session that's open to the university community, or in large ways, like students working directly with local organizations.

Another approach is to decrease the convenience, appeal, or utility of using AI. Strategies aligned with this approach include:

- Talking with students about how learning works can help students understand why certain uses of AI are unhelpful. For example, a student who knows the purpose of an assignment can think about which uses of AI might undermine that purpose.

- When students know that a task is a step toward or component of another assignment or assessment where AI collaboration will not be an option (e.g. an in-class presentation, exam, or interview), using AI for the intermediate steps is less appealing.

- Asking students to document, annotate, reflect on, or otherwise capture in metacommentary the process they used to complete an assignment makes AI use less convenient.

- AI will be less helpful when students are asked to draw on knowledge specific to class interactions that AI doesn't have context for. For example, AI doesn't know what case studies were discussed in class, or what the class' consensus about a particular discussion topic was.

- Having students complete some assignments in class can limit access to AI. Having students complete portions of assignments in class can also make AI use outside of class less appealing, as students have already started the relevant work.

- When students' work will be seen or used for something outside of class, there are more people to whom they are accountable.

- Showing students examples of ways AI can go wrong (like hallucinations or instances of algorithmic bias) can make it less appealing to use it uncritically. Evaluating these kinds of examples as a class can make this point particularly vivid.

- Taking time for honest conversations about why academic integrity matters can make the honor code feel less arbitrary and encourage accountability.

Exactly what these strategies look like in your courses and how many of them are feasible depends on your course context. Some might even be strategies you're already using. If you have questions or would like our help brainstorming possibilities, including questions about how these strategies might be applied in your discipline, we welcome you to reach out to us at ctl@pacific.edu. We can also point you toward further resources about some of the principles behind these strategies, like promoting metacognition, transparency in teaching, and co-creating class expectations.

Having offered some strategies you can use to discourage uncritical, dishonest, or otherwise inappropriate AI use, it's worth commenting on one approach that the CTL does not recommend. When you want to know that students worked in an AI-unassisted way, it's tempting to turn to AI detectors, in much the same way you might use a plagiarism detector to discourage copying from other students or online sources. However, there are three reasons we do not recommend these AI detection tools. First, the accuracy of these tools varies considerably[3] and even the most accurate can be beaten with sophisticated prompting.[4]

[3] This paper includes a review of the accuracy of six major AI detectors.
[4] For example, here's a paper discussing the "recursive paraphrasing" strategy for avoiding detection.

Second, at the same time that companies sell AI detectors to institutions of higher education, students are also being sold products for detection-avoidance. The CTL would rather that students and institutions use that money for more productive purposes. Finally, the errors that AI detectors make are distributed in inequitable ways. One concern is that students who can afford the paid versions of AI tools are more likely to avoid detection. So, the brunt of enforcement tends to fall on less financially advantaged students. Another concern is that detection tools are more likely to falsely flag a student's authentic prose as AI-generated when that student is not a native speaker of the language they're writing in. As a result, the brunt of false accusations tends to fall on these students.

Communicating Expectations

Whatever you decide about which uses of AI are and are not compatible with your course goals, students will need to understand those expectations. One way you can facilitate that is by adopting syllabus language about AI (you can find the CTL's suggestions about that here). Still, there are a number of further steps you can take to help make sure that you and your students are on the same page about these matters.

Because different faculty adopt different AI policies, students are likely to get some mixed messaging about what's appropriate. Moreover, as AI gets increasingly incorporated into the user interfaces of other technologies (e.g. writing assistants built into word processors), students may not always know what counts as using AI in the first place. As a result, one helpful way of supplementing your syllabus language is to make sure that your policy and its implications are clear to students. You can help ensure that students have read the policy by including an assignment in which students follow up on your syllabus, like by answering questions about it, introducing themselves, or starting a conversation about what they hope to get out of the course or what support they think they might need. Once students are familiar with the policy, a good first step is creating space for them to ask questions about it, and all the better if you can do this in a way that presupposes that students will have questions (e.g. moving from "any questions?" to "what questions do you have?"). You can help students identify areas of confusion by asking them questions about how the policy applies to hypothetical examples. Trying to apply the policy will help students think concretely about what it means, and their judgments about examples will help you gauge their understanding.

Even if students understand a policy when reviewing the syllabus, it might not be salient to them at the time that they're working on assignments for your course. To help reinforce your policies at appropriate times, you might try including reminders about the policy along with instructions for particular assignments, perhaps with additional context about how the policy applies to that assignment. Another option is to include following the policy as part of the

evaluation criteria for an assignment. For example, if your syllabus language asks students to document how they used AI, you might make that documentation procedure part of your grading criteria for assignments as well. If you have students do any metacognitive reflection about their work (e.g. writing briefly about how they approached an assignment and what they might do differently next time) you can also ask them about what they did to ensure that their AI use was compatible with course policies. Honor pledges that students complete when turning in assignments (e.g. signing a form asserting that the AI policy has been followed) can also help to ensure that the policy is on students' minds when its most relevant to their work.

Any policy – whether related to AI or not – works best when students understand and appreciate the reasons for it. They may not be motivated to follow a policy if it feels arbitrary or unjustified. So, another way of supplementing your syllabus language is to get students' buy-in. A good first step is explaining the reasons behind the policy. Going further, you can gauge students' feelings about the policy, either in class or through an outside of class poll (maybe they're anxious about being accused of cheating, maybe they're concerned about other students cheating, maybe they're intimidated by the type of work the class will require). Showing that you understand their perspectives and have taken them into account will help to build trust. An extension of this strategy is giving students a role in the creation of the policy itself. For example, you might come up with a few possibilities you'd be comfortable with, then let the class choose. Alternatively, you might lay out the considerations you think are relevant to a policy that will work for your class, then develop one in conversation with students.

Conclusion

Both for those who are excited about AI and those who are wary of it, thinking about pedagogy and AI can feel overwhelming. There's a lot you might want students to know, you might be more or less intrinsically interested in following new developments in this technology, and time and attention spent on AI might compete with time and attention spent on activities and assignments that are more directly related to your course's content. We hope that this document (and the workshop series that inspired it) helps to make things more tractable by providing a place to start.

As you make decisions about your own teaching, here are three closing thoughts that might help to keep that process manageable. First, you don't need to implement all of your ideas at once. It might be more sustainable to choose one or two of the strategies suggested above – or however many feels doable – to try out, then build up incrementally from there. Second, it can be reassuring to consider ways that your approach to AI might be congruent with approaches to teaching you are already using. Things like reflecting on the pedagogical purpose of your assignments, being transparent with students about why they're doing what they're

doing, discussing how learning works and how to engage in your course most productively, working toward buy-in for course policies and norms, striving for assignments that are authentically rewarding to work on, and creating space for students to reflect on their work and what they're taking from it are all worthy pursuits independently of their application to AI. Viewed in this way, you might find that many of the things you decide on when doing course design with AI in mind are extensions of things you are already doing, or already interested in trying. Finally, pedagogical decisions are always a work in progress. We are all learning alongside our students, and that's okay.